

Assignment-3

1. An RDD has a data size of 1.2 GB stored in HDFS with a block size of 128 MB. Calculate the number of partitions created by Spark. If an executor has 4 cores, how many tasks can run in parallel?
2. A Spark cluster has 5 worker nodes with 16 cores per node. If the driver uses 2 cores and each executor is configured with 4 cores, calculate the total number of executors and the number of parallel tasks.
3. An executor is allocated 10 GB of memory. If spark memory fraction is 0.6 and spark memory storage Fraction is 0.5, calculate the usable memory, storage memory, and execution memory.
4. A Spark job processes 600 GB of input data using 300 mappers and 100 reducers. Calculate the amount of data processed by each mapper and each reducer.
5. A Spark job consists of 200 tasks. Each task takes 5 seconds to execute, and the cluster has 10 executor cores. Estimate the total job execution time.
6. A dataset is divided into 8 partitions with a replication factor of 2. Calculate the total number of data copies and explain whether the job can continue if one copy of a partition fails.
7. Out of 120 tasks, 90 are data-local tasks taking 3 seconds each and 30 are non-local tasks taking 6 seconds each. Calculate the total execution time and the approximate execution time if the cluster has 8 cores.